

Training-time electrical elimination in physics-informed neural networks for electrothermal phase reconstruction

Authors: [author names and affiliations to be supplied]

Corresponding author: [name and email to be supplied]

Abstract

A final electrical solve can repair terminal outputs while leaving inaccurate internal states unchanged. We examine how this constraint should participate in learning temperature and phase in a synthetic two-dimensional electrothermal device. A partially eliminated PINN solves the grounded electrical problem during training and uses the same face resistances for local Joule deposition. Soft PINN and interpolation controls receive the same final electrical solve. Four paired full-label fits on two separately reconstructed protocols retain the complete device-advantage criterion under three numerical references and two readers; current and power error reductions over the original F controls are 47.03-76.52% and 47.86-77.50% with the spatial reference and finer reader. Against a soft control with mixed-measure space-time coverage enhancement, the eliminated configuration reduces current error by 52.68% and 66.55% for the two shorter-protocol initializations under the spatially refined reference and finer electrical reader. Strong interpolation supplies a ranking counterexample, and matched controls do not establish added reconstruction value from the thermal/phase interior residual package. A complete second-cycle phase gap and subsequent bounded phase-correction tests retain this limitation. In a common-measure diagnostic, an observation-preserving neural correction lowers its scalar physical objective by 89.91% while increasing the raw phase-residual square 10.32-fold. Thus objective reduction and repaired ports cannot substitute for separately evaluated dynamic consistency. The results support a tested electrical-interface configuration and delimit its attribution; they do not establish neural-specific phase completion, strict two-cycle usability, formal out-of-distribution generalization or material validation.

Keywords: physics-informed neural network; electrothermal coupling; electrical elimination; missing phase observations; phase-field reconstruction

1. Problem and information conditions

Small potential errors can produce large contact-current and local-heating errors in a coupled phase-transition calculation. Electrical repair can restore terminal consistency without repairing the learned evolving state. We therefore ask two separate questions: whether training with the electrical constraint improves reconstruction after every method receives the same final solve, and whether the additional thermal/phase interior residuals contribute under complete and phase-gap observation conditions.

PINNs combine observations with explicit differential-equation residuals [1]. We study how electrical constraints participate in reconstructing transient temperature and phase from prescribed field observations. The task is an offline, known-model reconstruction benchmark: geometry, coefficients, initial and boundary conditions, and the complete drive are supplied. Each of the two protocols is fitted separately. Its original observation set contains 28,875 positive-time locations and 86,625 noiseless scalar labels for potential, temperature and phase. These locations constitute 1.8046875% of the $3,200 \times 500$ positive-time carrier positions; at each observation time, 231 spatial locations cover 7.21875% of the carrier cells. Noiseless labels still inherit the carrier's numerical discretization error. These experiments do not establish a minimal sensor arrangement, zero-shot transfer or calibrated material reconstruction.

The contribution is a specific coupling interface and a matched test of its reconstruction benefit. Neural temperature and phase determine a grounded finite-volume electrical solution. The same face resistances determine local Joule deposition in the thermal balance. Providing the same final electrical solution to the soft and interpolation controls tests whether final electrical repair explains the measured differences. The existing experiments support this training configuration relative to the tested soft comparator. They do not establish a new implicit-differentiation principle or an independent benefit from the additional thermal/phase interior residuals.

Differentiable constrained learning already includes solver-in-the-loop correction [3], hybrid FEM-NN models [4], PDE-constrained layers [15], and modular implicit differentiation [5]. Related approaches enforce hard boundary maps or constrained optimization [16], prescribed integral targets [12], differential-algebraic constraints [17], and

local discrete projections [18]. Here the transient T/ϕ states remain neural, while the full grounded electrical subproblem is eliminated. This identifies a physical interface adaptation and its controlled comparison; it does not imply general superiority over all penalty or constrained-optimization methods.

Reduced and all-at-once PDE formulations provide the broader setting for eliminating a constrained variable versus jointly optimizing all fields [13]. Here only quasi-static potential is eliminated; the transient states remain neural.

The field network follows the smooth modified-MLP construction of Wang et al. [2]. The spatial cell-balance thermal residual is related to control-volume learning [6], while the nonconserved phase dynamics use an Allen-Cahn-type potential [7]. The device layout and electrothermal feedback are inspired by wall-cell phase-change-memory modeling [8]; its Ge-rich GST material formulation is distinct from the present dimensionless single-phase variable. The parameters below define the synthetic benchmark and are not an oxide calibration.

2. Two-dimensional physical object

2.1 Equations, geometry and known coefficients

The computational cell is $\Omega = [-1, 1] \times [0, 1]$, with coordinates x and z and time $0 \leq t \leq 2.5$. Potential V , reduced temperature T and phase fraction ϕ obey

$$\nabla \cdot (\sigma(T, \phi) \nabla V) = 0. \quad (1)$$

$$T_t + L\phi_t = \alpha \Delta T - \gamma T + Q\sigma(T, \phi) |\nabla V|^2. \quad (2)$$

$$\phi_t = M(T) [\varepsilon^2 \Delta \phi - \partial_\phi W(\phi, T)]. \quad (3)$$

The conductivity, mobility and local phase potential are

$$\sigma = \exp[\beta T + \log(R)\phi^2(3 - 2\phi)], \quad (4)$$

$$M(T) = m_c + (m_h - m_c) \text{sigmoid}[(T - T_c)/w_M], \quad (5)$$

$$W = B\phi^2(1 - \phi)^2 + D(T_c - T)\phi^2(3 - 2\phi). \quad (6)$$

We use $\alpha = 0.10$, $\gamma = 4$, $L = 0.05$, $Q = 4$, $\beta = 0.25$, $R = 8$, $\varepsilon = 0.04$, $B = 1$, $D = 6$, $T_c = 0.45$, $m_c = 0.5$, $m_h = 5$ and $w_M = 0.08$. Mobility multiplies the entire phase bracket; it is not moved inside a divergence. Because W depends on the externally evolving temperature, the model is not assigned an autonomous monotone free-energy constraint.

The full top electrode is driven by $U(t)$; a centered bottom segment $|x| \leq 0.35$ is grounded. All remaining electrical faces are insulating. The top temperature is zero; the other thermal boundaries satisfy $\partial_n T + 0.25T = 0$. Phase satisfies $\partial_n \phi = 0$ on every boundary. Initially $T = 0$ and

$$\phi_0 = 0.02 + 0.01 \exp \left[-\frac{1}{2} \left((x/0.18)^2 + ((z - 0.12)/0.10)^2 \right) \right]. \quad (7)$$

These assumptions define a synthetic wall-cell benchmark. No dimensional material parameters, experimental memory operation, or oxide-specific constitutive identification are inferred from it.

2.2 Initial-boundary interpretation

For the initial phase in Eq. (7), at $x=0$ on the bottom boundary, its outward normal derivative is $-0.12 \exp(-0.72) = -0.0584102707151966$. It is therefore not pointwise compatible with the phase Neumann condition at $t=0$. The reference applies the stated discrete no-flux operator during subsequent evolution; the neural representation retains the exact initial map and penalizes phase boundary residuals. We do not assume classical smooth compatibility on the closed space-time boundary.

Direct inspection of the twelve saved historical full-label calibration/L-BFGS/audit pools for the two protocols and two seeds finds no $t=0$ boundary node and hence zero boundary quadrature mass there. This finite-pool fact is distinct from a probabilistic argument about continuous random draws. The existing Adam rule generates boundary times from the four physical windows without explicitly inserting $t=0$; it does not mathematically exclude a zero random variate. The compatibility limitation is retained regardless. Its effect on the E/D_E ranking is unknown; neither the initial state nor the boundary rule has been changed.

2.3 Pulse histories and observations

Each pulse rises linearly from zero to 0.72 over its first 0.05 time units, holds until local time 0.27, then falls to zero at 0.35. The original protocol starts two pulses at 0 and 1.25. The intervention changes only the second start to 1.01, leaving the pulse shape, total time, coefficients and boundaries fixed. The input contains exactly two pulses; it is not periodically extended to create a third pulse. Advancing the second pulse by 0.24 preserves its phase relative to the 0.02 observation spacing.

Support trajectories use 80×40 cells, a main time step of 0.0025 and saving every two steps. The spatial and temporal masks take indices 0,4,8,... and the last index, fixed independently of field values. The resulting $21 \times 11 \times 126$ mask contains 231 analytic initial positions and 28,875 positive-time positions, with 86,625 scalar field labels. Training receives only these samples, coordinates and known physical laws. Dense support fields, reference fields, reference currents or heating, and sealed stress cases do not enter these training losses or within-run endpoint selection. Published reference evaluations informed subsequent research questions and interpretation; the overall development process is not claimed to be reference-blind.

The evaluation reference uses 160×80 cells, a main time step of 0.000625 and saving every four steps, giving 1001 times. Both numerical trajectories use backward-Euler finite-volume steps with converged electrical/phase/thermal block iterations and logit Newton for phase, as specified in Supplement S1.1. Their algebraic convergence tolerances do not certify discretization accuracy. The grids serve different roles; their existence alone is not a convergence study. Each protocol has its own observations and is fitted separately. The intervention is consequently a complete-case adaptation experiment, not zero-shot transfer or a formal out-of-distribution test. The two seed values are reused across protocols for paired reporting, not counted as four independent physical cases.

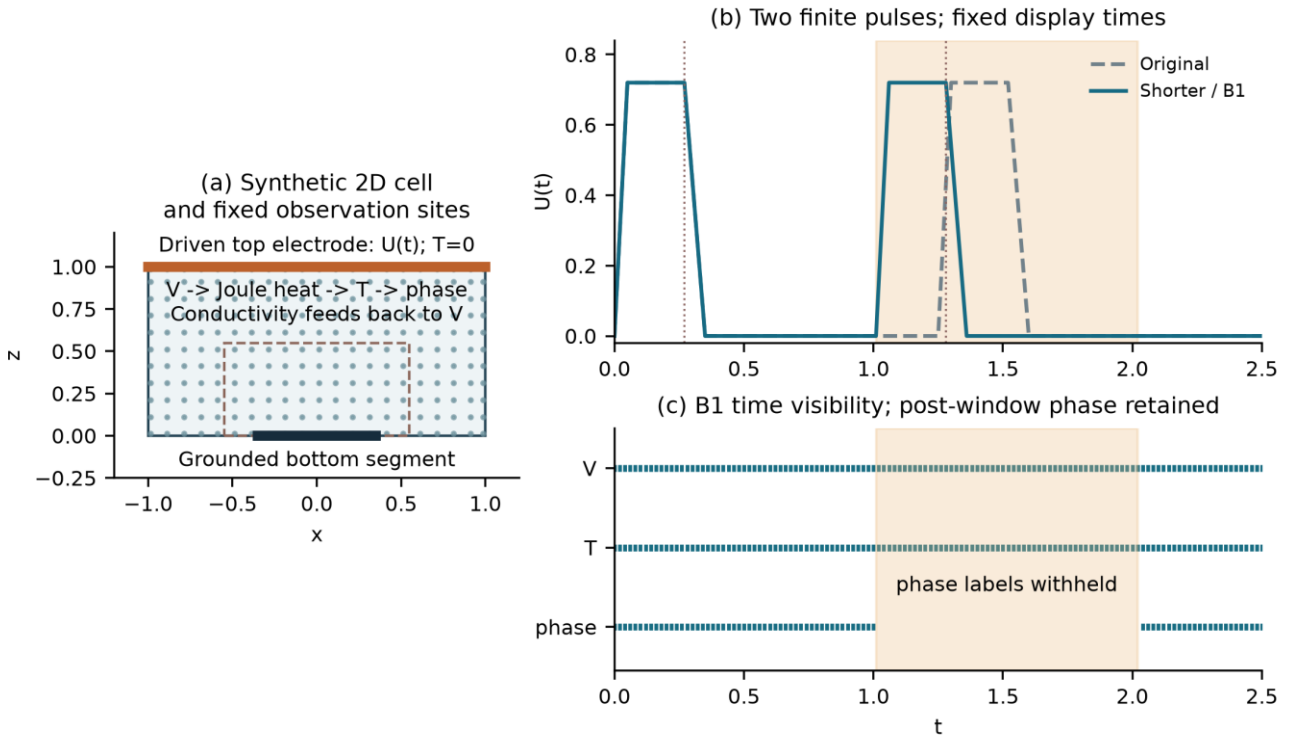


Figure 1. Actual geometry, fixed spatial sampling and finite pulse protocols. The shaded interval $W=[1.01, 2.02]$ removes only phase labels in B1; potential and temperature remain observed. Phase observations after W remain available. Dashed vertical lines mark the predeclared display times 0.27 and 1.28. The dashed spatial box is the fixed state-error ROI.

3. Method and matched controls

3.1 Neural states and electrical elimination

Let $u_\theta = (T_\theta, \phi_\theta)$. Independent smooth modified-MLP heads [2] have width 64 and four hidden layers; the shared temperature architecture also includes a small additive latent adapter described in the Supplement. FP64 arithmetic and identical output mappings are used in both methods. With $a(t) = 1 - \exp(-t/0.35)$,

$$T_\theta = 2.5a(t)(1 - z)\text{sigmoid}(h_T), \quad (8)$$

$$\phi_\theta = \text{sigmoid}[\text{logit}(\phi_0) + 8a(t)h_\phi]. \quad (9)$$

The eliminated method, E, evaluates σ_θ from these states and solves the grounded electrical system

$$A(\sigma_\theta)v^* = f(U, \sigma_\theta). \quad (10)$$

For an internal face e between cells i and j , the half-cell distances d and face area A_e define

$$R_{ei} = d_{ei}/(\sigma_i A_e), \quad R_{ej} = d_{ej}/(\sigma_j A_e), \quad g_e = (R_{ei} + R_{ej})^{-1}. \quad (11)$$

Electrode faces use the corresponding half-cell resistance and the actual heater overlap. Insulating faces contribute no outward conductance. Positive conductances on the connected grid, with Dirichlet electrodes fixing its gauge, make the assembled electrical matrix nonsingular. Electrical enforcement means solving this discrete system to numerical precision; it is not an exact continuum solution. The implementation uses sparse factorization, not a dense inverse. For a scalar loss, the complete first-order derivative is obtained from

$$A^T z = \partial \mathcal{L} / \partial v, \quad d\mathcal{L} = d\mathcal{L}_{\text{direct}} + z^T(df - dA v^*). \quad (12)$$

The direct term includes the explicit dependence of heating on conductivity. The right-hand side f also depends on electrode conductance, and its derivative is retained. A factorization may be reused for the adjoint of the same forward solve, but not after changing the state. This is an application of implicit differentiation [5], with no claim of a new differentiation theorem or support for arbitrary second-order derivatives.

3.2 Consistent local Joule deposition and remaining residuals

Given either the solved or neural potential, face current and deposition are

$$I_e = g_e(v_i - v_j), \quad P_{e \rightarrow i} = I_e^2 R_{ei}, \quad P_{e \rightarrow j} = I_e^2 R_{ej}. \quad (13)$$

Electrode power enters the neighboring boundary cell. For cell volume ω_i , q_i is its deposited power divided by ω_i . This allocation is not generally an equal split between adjacent cells. It gives

$$\sum_i \omega_i q_i = P_J, \quad (14)$$

where P_J sums all internal and electrode dissipation. Equation (14) is an accounting identity for the shared face network, even when a neural potential violates electrical balance. Equality with terminal power UI additionally requires the electrical equations. Neither identity by itself establishes accuracy relative to the reference heating field. P_J denotes electrical dissipation; the reduced thermal source is Qq , so its integral is QP_J . This normalization and the temperature-dependent phase potential do not imply a general total-free-energy decay theorem.

The temperature residual uses a cell balance,

$$R_{T,i} = \partial_t(\bar{T}_i + L\bar{\phi}_i) + \gamma\bar{T}_i - \frac{\alpha}{\omega_i} \sum_f A_f (\nabla T_\theta)_f \cdot n_{if} - Qq_i. \quad (15)$$

Cell means use midpoint quadrature. Temperature gradients are differentiated at shared face nodes with opposite normals for neighboring cells, including the model's own gradients on boundary faces. The same thermal boundary residual remains in both losses. The factor Q is applied once. The phase residual is

$$R_\phi = \phi_t - M(T)[\varepsilon^2 \Delta \phi - 2B\phi(1 - \phi)(1 - 2\phi) - 6D(T_c - T)\phi(1 - \phi)]. \quad (16)$$

Thus the thermal interface is a spatial cell-balance residual and the phase interface is a pointwise AD residual. They are not claimed to equal the previous pointwise thermal formulation or the full space-time construction of [6]. At $U = 0$, E uses the analytic electrical solution $v = q = 0$ while thermal and phase evolution remain active.

3.3 Soft comparator and training objectives

The soft electrical PINN, F, trains all three field heads. Its potential is $V_\theta = U(t) \exp(h_V)/[\exp(h_V)+1-z]$, preserving the voltage range and the exact top value. It evaluates the same face operator and Joule deposition as E, but penalizes

$$r_{e,i} = (A(\sigma_\theta)V_\theta - f(U, \sigma_\theta))_i/\omega_i. \quad (17)$$

No additional, differently averaged electric boundary loss is added: electrical boundary conductances already enter this residual. Let L_{obs} denote the common weighted observation loss, including the full initial-logit increment for phase, and let L_{BC} and L_{IC} be the common remaining boundary and initial losses. Their exact measures are specified in Supplement S2. With one calibration a, b at the shared parent and λ rising to 0.1,

$$J_{T\phi} = \frac{1}{3}\mathbb{E}_\rho[(R_T/4)^2 + (R_\phi/5)^2], \quad (18)$$

$$\mathcal{L}_E = L_{\text{obs}, E}/a + \lambda(5L_{\text{BC}} + L_{\text{IC}} + J_{T\phi, E})/b, \quad (19)$$

$$\mathcal{L}_F = L_{\text{obs}, F}/a + \lambda(5L_{\text{BC}} + L_{\text{IC}} + J_{T\phi, F})/b + \frac{\lambda\eta}{3b}\mathbb{E}_\rho[r_e^2]. \quad (20)$$

Here ρ is the uniform space-time target with the declared time-window quadrature. Scales a and b are shared within a pair and frozen, with a numerical floor of 10^{-12} . The primary F uses $\eta = 1$. A development control uses a one-time gradient-balanced η ; a second control replaces the 128 sampled electrical cells by the volume mean over all 3200 cells at each physical time. The latter changes spatial integration, not the penalty by a factor of 25. Thermal and phase terms keep their original interfaces.

E freezes the unused potential head. Its voltage observations differentiate through conductivity and the solve, whereas F voltage observations act on its potential head. E enforces its constraint at powered observation and physics times; F penalizes it at physical residual times. Initial potential, active parameters, constraint strength and gradient maps consequently differ. These are explicit parts of the method-package comparison, not hidden evidence for an isolated VJP mechanism.

3.4 The D_E ablation and strong interpolant

E learns T and ϕ through the electrical solve and includes the thermal/phase interior residuals, boundary terms and initial constraints. F retains a neural potential and the frozen finite electrical penalty; its endpoint receives the identical electrical readout used by E . D_E shares E 's neural state mapping, temperature adapter, electrical solve, complete first-order conductivity feedback, voltage-observation loss, boundary/initial constraints, parent, calibration and optimization limits; only the thermal/phase interior package is removed. D_E is consequently a physics-containing ablation, not a data-only network.

B_E uses time-PCHIP and spatial-linear interpolation of the supplied T/ϕ observations, the original initial-logit restoration and known boundary extensions, followed by the same electrical solve. It is constructed once per protocol/information condition and does not use the V labels. The E/B_E comparison therefore measures practical competitiveness under the stated input utilization, not a strictly identical information-use intervention.

The precise ablation objective is

$$\mathcal{L}_{D_E} = L_{\text{obs}, E}/a + \lambda(5L_{\text{BC}} + L_{\text{IC}})/b. \quad (21)$$

The scale b is still calibrated from the complete parent physical package. Thus E minus D_E removes exactly the interior term rather than renormalizing the shared boundary or observation contribution. Targeted value/gradient checks and full-visible compatibility checks precede the new execution.

The controls answer distinct questions. F_{cov} tests the declared mixed-measure strengthening of the soft electrical configuration; D_E tests the additional thermal/phase interior-residual package; B_E tests competitiveness under its stated use of visible fields; $B1$ tests a complete second-cycle phase-observation gap. E/F and E/F_{cov} remain configuration comparisons and do not isolate the VJP.

3.5 Common parents, fixed endpoints and readouts

For each protocol, seeds 29 and 43 initialize fresh networks and a zero-output adapter, without loading trained weights, optimizer history or calibration. A common observation-only parent receives 2400 Adam updates [9] and 600 full-observation L-BFGS evaluations, 200 per field head [10]. E and F then start from that same fitted parent with fresh optimizers. Each branch receives 1500 Adam updates and at most 300 fixed complete-objective/gradient evaluations. Every line-search trial and repeated evaluation counts; budget termination restores the last accepted state. No reference score selects a checkpoint.

The learning rate, Adam parameters, clipping, observation stream, physical time windows and residual scales are shared. L-BFGS uses fixed data and quadrature, without resampling or gradient clipping inside its closure. Equal update and evaluation caps do not imply equal computation: E incurs sparse forward and adjoint solves, while F differentiates a potential network and its explicit residual. Actual work is retained in Supplement S6; no speedup is claimed.

The shorter-protocol D_E controls share the corresponding full-label parents. B1 instead establishes two entirely new observation-only parents with only its visible phase labels, then runs E, D_E and F in each seed without selecting a favorable checkpoint or skipping an arm based on performance. All fixed endpoints are locked before reference scoring. The common 160 by 80 and 240 by 120 electrical readers evaluate every model at the same 1001 times. Fine-grid T/phase are direct queries of the same locked function. T/phase and event scores remain on 160 by 80; only fine potential is conservatively restricted for the inherited field guard, while native currents, power and Joule deposition stay native.

3.6 Distinct error and event measures

The historical primary comparison uses the same 160×80 electrical layer at the reference times. The 240×120 reader changes prediction-side electrical discretization while preserving the historical scoring measure. The unprojected F readout is retained separately. For a fixed F, projection changes V, q and electrical readouts while leaving T, ϕ and every phase event unchanged. The error measures distinguish these effects.

S is the space-time mean symmetric difference of active sets, where active means $\phi \geq 0.5$; it uses the full cell domain and global trapezoidal time weights. Raw E_ϕ is phase RMS on the ROI $|x| \leq 0.55$, $0 \leq z \leq 0.55$. E_T is ROI temperature RMS divided by 0.45, and E_V is unnormalized full-domain potential RMS. Device errors are

$$\mathcal{E}_I = \frac{\|I - I_{\text{ref}}\|_t}{\|I_{\text{ref}}\|_t}, \quad \mathcal{E}_P = \frac{\|P_J - P_{J,\text{ref}}\|_t}{\|P_{J,\text{ref}}\|_t}, \quad \mathcal{E}_W = \frac{|\int (P_J - P_{J,\text{ref}}) dt|}{|\int P_{J,\text{ref}} dt|}. \quad (22)$$

The time norm is trapezoidal RMS over the complete interval. The matched device criterion compares predicted bottom current with the reference top-current trace, retaining the historical scoring convention. The top-current noninferiority guard uses the predicted top current. For projected states these terminal traces agree to solve accuracy; the archive also reports bottom-native errors as a separate diagnostic. Unprojected soft currents can disagree substantially and must not be interchanged. Percentages below are 100 times normalized errors, not dimensional percentage errors in temperature or phase. All absolute, percentage-point and relative differences are provided in the unified tables.

For the same heating-window measure, let M_{pred} , M_{ref} and M_{TP} be predicted, reference and overlapping active masses. When both masses exceed the numerical denominator floor, the definitions give

$$\text{recall} = \text{precision} \frac{M_{\text{pred}}}{M_{\text{ref}}} \leq \min \left(1, \frac{M_{\text{pred}}}{M_{\text{ref}}} \right). \quad (23)$$

This identity separates an insufficient predicted active mass from mismatch in space and time at a fixed mass. It is a descriptive support diagnostic, not a new loss or evidence that a particular neural mechanism caused the error. The original near-zero floors remain in the scorer. The strict gates are project-defined evaluation requirements, not experimental device-reliability standards or estimates of a success probability.

Predeclared phase-reconstruction and device-function criteria require 10% improvements in their named primary metrics with 5% noninferiority guards and fixed absolute floors. They are retained in Supplement S3, rather than used as a substitute for reporting error magnitudes. Failure to cross an advantage threshold does not demonstrate equivalence. Strict event usability is separate: both cycles must meet onset error ≤ 0.005 , recall ≥ 0.9 , precision ≥ 0.8 , mass ratio 0.8-1.2, and the specified peak, locality and recovery conditions. An onset is the linearly interpolated upward crossing of ROI active fraction 0.02. Support metrics integrate cell-volume overlap during each heating window.

For projected states, $P_J = UI$ to electrical solve accuracy. Current and power errors thus differ in drive-dependent temporal weighting, but are not independent physical replications. Likewise, a small integrated energy error may reflect cancellation. We therefore retain the signed per-pulse errors, event metrics and local state errors even when global device criteria pass.

3.7 Frozen second-cycle phase-gap experiment

The new observation condition removes phase labels on the complete second cycle $W=[1.01,2.02]$ of the existing shorter protocol. V/T observations and phase outside W , including $t>2.02$, remain available. The exported data contain 17,094 positive-time phase labels and omit 11,781; V and T each retain 28,875. The original observed grid contains 1.02 through 2.02 within W , with no observed node at 1.01. The last preceding and first following visible times are 1.00 and 2.04. The analytic initial condition remains known.

Two new observation-only parents, seeds 29 and 43, each initialize E, D_E and F under the frozen recipe. No complete-label trained state, calibration or interface pool is reused. Phase observation mass restricts the original coordinate quadrature before normalization. A threshold-straddling cell is admitted only when all its original adjacent corners remain visible. The primary question is whether E adds information over D_E in W , under the separately named A_w criterion. Whole-history A/B, strict events, outside-window costs and E/F/B_E comparisons remain separate. This is offline reconstruction of a developed protocol, not online forecasting or a wholly unobserved phase trajectory.

A_w applies the original phase-advantage formula to W with its own time weights and normalizers: each of the S and phase-RMS improvements must satisfy

$$E_{\text{control}} - E_{\text{candidate}} \geq \max(0.1E_{\text{control}}, \epsilon_{\text{abs}}).$$

The absolute floor is the corresponding inherited tolerance. T, top-current and potential retain their original 5% noninferiority guards. The bottom-current convention remains separately reported. The outside measure integrates $[0,1.01]$ and $[2.02,2.5]$ separately, without bridging W . Its unnormalized integrals plus those of W reproduce the full-history integrals. Outside costs include all seven named state/device quantities. Original full-history A/B and strict-event rules remain unchanged. This is a complete withheld time window for phase, with V/T still observed inside it; it is not a formal out-of-distribution or forecasting test.

4. Reconstruction after common electrical repair

4.1 Four paired full-label comparisons

All four clean E/F pairs retain the complete device criterion under every existing reference and both readers. Under the spatial reference with the finer reader, current-error reductions are 47.03-76.52% (0.78-2.28 percentage points) and power-error reductions are 47.86-77.50% (0.82-2.39 points). The comparison therefore survives giving F the same final electrical solve. Four pairs on two related protocols are the comparison units; reference, reader and saved-time counts do not increase that sample size.

Table 1. Complete full-label objects under the spatial reference and fine reader. Phase uses the original 160 by 80 ROI; native device NRMSE values are percentages. B_E is one deterministic object per protocol. D_E is shown alongside the learned controls rather than introduced only after the main results.

Protocol	Seed	Role	Phase RMS	Current (%)	Power (%)
original	29	E	0.0210621	0.883264	0.889983
original	29	F	0.0247498	1.66733	1.707
original	43	E	0.0224575	1.59785	1.62156
original	43	F	0.0251648	3.07484	3.18207
original	shared	B_E	0.0233096	1.61121	1.63659
shorter	29	E	0.0199363	1.01558	1.0237
shorter	29	F	0.0244496	2.27681	2.34323
shorter	43	E	0.0197631	0.70046	0.694693
shorter	43	F	0.0240993	2.98378	3.08711
shorter	shared	B_E	0.0252168	1.9043	1.9475
shorter	29	D_E	0.0194683	0.958122	0.965194
shorter	43	D_E	0.0197085	0.642947	0.624503

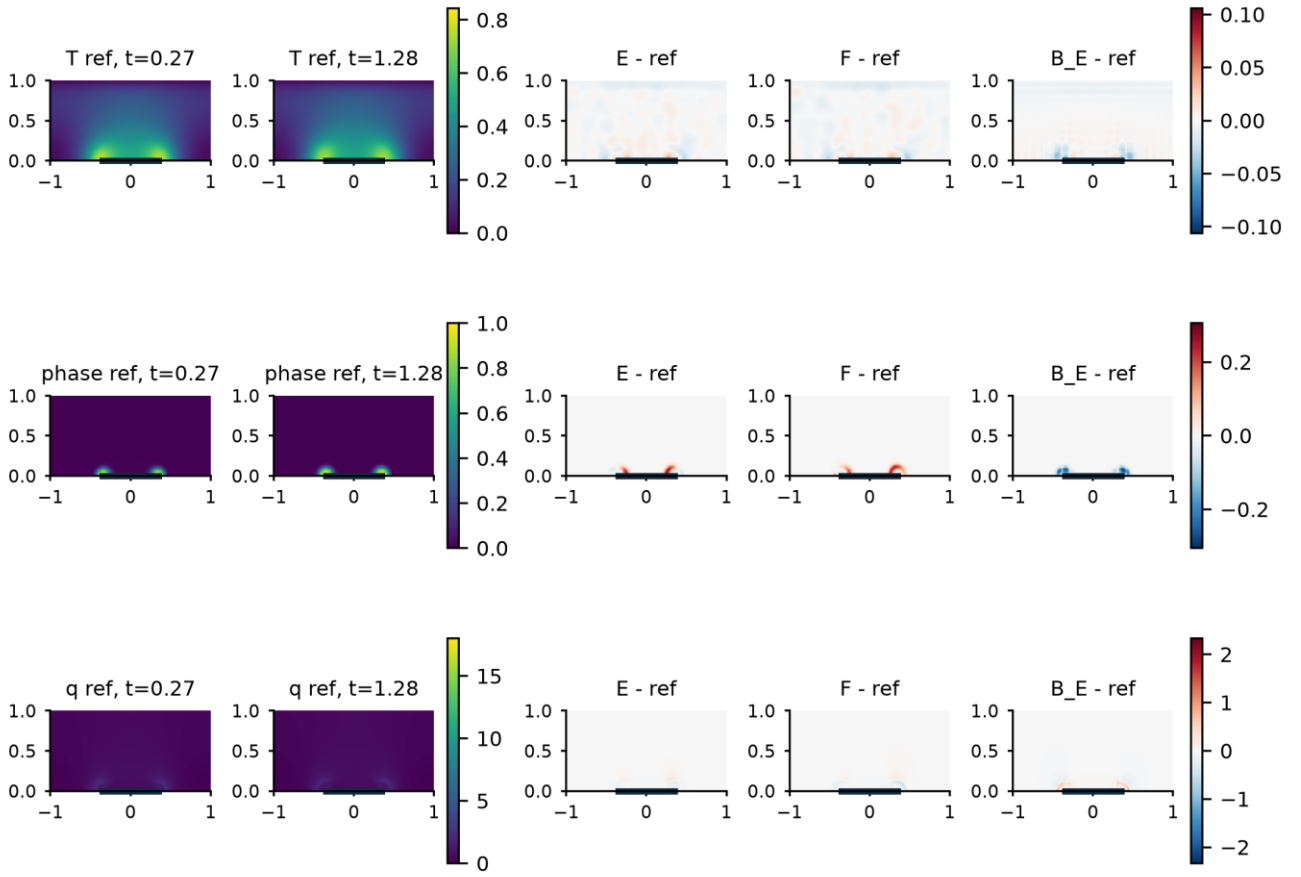


Figure 2. Native 240 by 120 physical states and signed errors for the full-label shorter protocol. The first two columns show spatial-reference T, phase and Joule density at the two predeclared plateau endpoints. The remaining columns show E, repaired F and B_E minus reference at t=1.28 for seed 29. Error color scales are shared with the seed-43 companion in the supplement. These physical panels are descriptive native-grid views; the primary T/phase/event measure stays on 160 by 80.

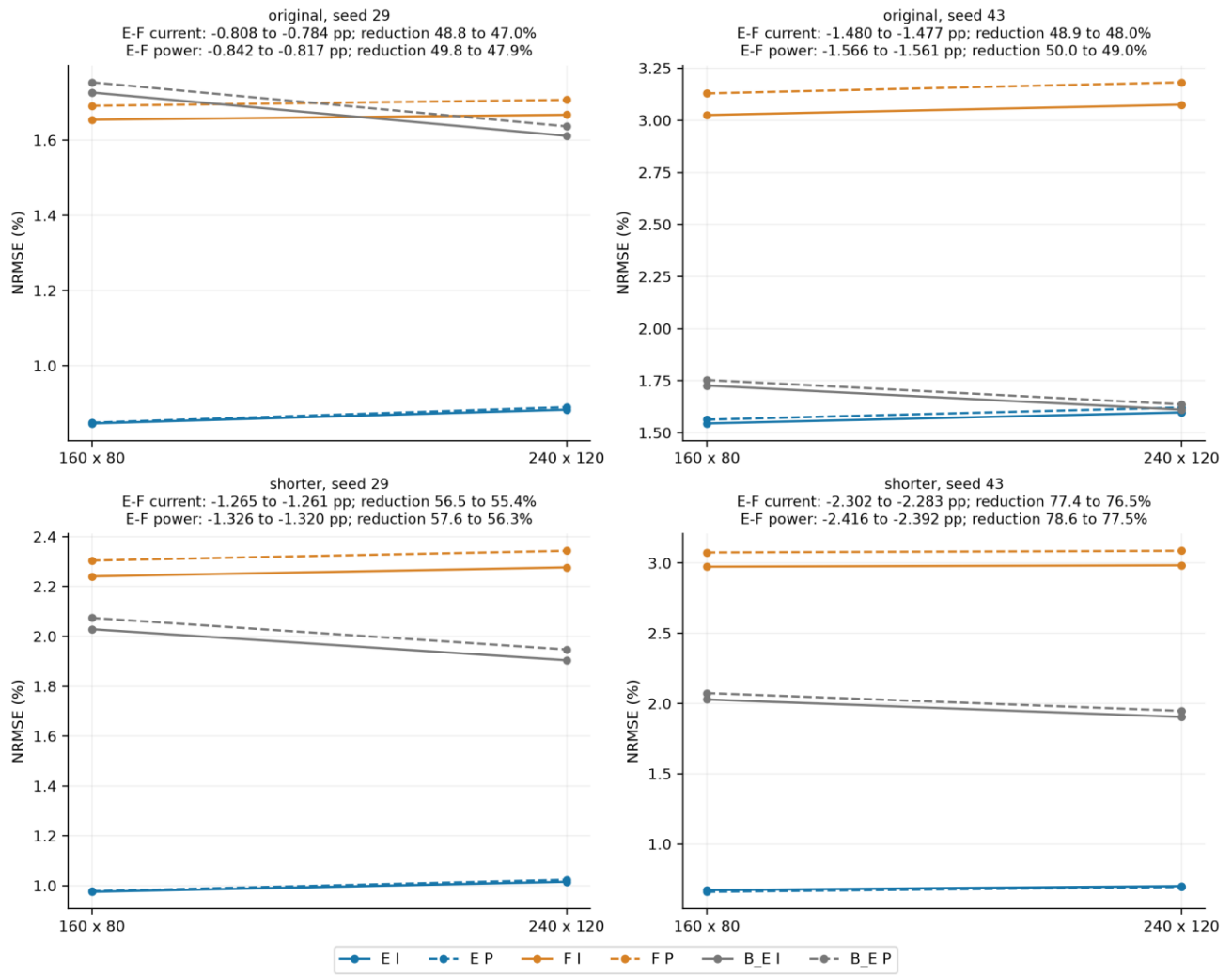


Figure 3. All four historical E/F pairs under the same native spatial reference and the two electrical readers. Solid and dashed curves show current and power NRMSE. B_E is repeated visually across seeds but is one baseline per protocol. Signed E-F differences use percentage points. The complete criterion also includes all field guards.

The development controls illustrate why repair is a necessary comparison: with T/phase fixed, projection reduces F_{raw} current NRMSE from 279.30% to 1.071% and power NRMSE from 20.118% to 1.065%, while E gives 0.694% and 0.682%. A full-spatial finite-penalty control does not remove that development difference, but its integrated energy error is better. These controls delimit the tested package; they are retained in Supplement S4 and are not pooled as clean replications.

4.2 The strong interpolation counterexample

This robustness does not extend uniformly to B_E. For original-protocol seed 43 and the 240×120 reader, the full-precision records give:

Table 2. Continuous interpolation counterexample for original-protocol seed 43 with the fine reader. Negative relative error reduction means larger E error; full-precision records determine the unchanged A/B decisions.

Reference	Current NRMSE: E / B E (%)	Power NRMSE: E / B E (%)	Relative current / power error reduction (%)	Original A / B
Original	1.631055 / 1.620006	1.657409 / 1.646429	-0.681976 / -0.666882	False / False
Time-refined	1.653271 / 1.570460	1.681856 / 1.596088	-5.273031 / -5.373634	False / False
Space-refined	1.597848 / 1.611207	1.621556 / 1.636592	0.829142 / 0.918751	False / False

Negative reduction denotes larger E error. Under the time-refined reference the differences are 0.082811 and 0.085768 percentage points, respectively. Under the spatial reference E remains slightly better, below the prescribed practical margin. The first two comparisons already failed B, so the continuous reversals introduce no additional Boolean decision changes. Full-precision source fields and line locations are preserved in [the P04 table](#); these display values do not re-adjudicate thresholds.

4.3 Mixed-measure coverage enhancement

Two additional soft-PINN continuations retain the short-interval full-label parents, observation streams, non-electrical objectives, calibration and soft-network Joule source. Their electrical penalty replaces the original term with equal normalized physical-time and powered-observation-time measures over all 3200 cells. This changes spatial coverage and temporal measure together; it is not a pure coverage or isolated VJP experiment.

Against F_cov, E satisfies the original whole-history phase criterion A and device criterion B in 12/12 conditions each: two initializations, each evaluated with three references and two readers. The twelve rows are sensitivity conditions of two endpoints, not independent repetitions. F_cov satisfies A/B against its historical F in 0/12 and 0/12 conditions, and satisfies the unchanged strict rule in 0/12. These are twelve sensitivity records for two new continuations, not twelve independent runs.

Table 2a. Continuous effects against F_cov under the spatially refined reference and finer electrical reader. Current NRMSE is reported in percent from the native 240 by 120 electrical reader; phase RMS uses the original 160 by 80 ROI. Values are display summaries of the existing spatial/fine/E_vs_F_cov rows, not new threshold decisions.

Seed	Current F cov (%)	Current E (%)	Reduction (%)	Phase F cov	Phase E	Reduction (%)
29	2.1463	1.01558	52.68	0.0241544	0.0199363	17.46
43	2.09424	0.70046	66.55	0.0244337	0.0197631	19.12

F_cov nevertheless reduces current and power error relative to F by 1.36-9.00% and 3.73-11.32% for seed 29, and 27.43-31.72% and 29.79-34.00% for seed 43. Its phase-set error S increases by 11.40-12.57% and 13.05-15.28%, respectively, violating the set-error noninferiority component in every condition. Thus the failed complete criteria coexist with continuous port improvements. Continuous metrics, all criterion failures and original network readouts are retained in Supplement S24. The result tests robustness to this stronger soft configuration; initialization, active-variable and enforcement differences still prevent single-mechanism attribution.

5. Dynamic residual increments and failure diagnostics

5.1 Full-label matched D_E evidence

In both clean shorter-protocol pairs, D_E has lower phase, current and power RMS errors than E under all three references and both readers. Neither pair establishes the prescribed added-residual phase or device increment. With the spatial reference and finer reader, E exceeds D_E in phase/current/power error by 2.404%/5.997%/6.061% for seed 29 and 0.277%/8.945%/11.239% for seed 43. Other outcomes are mixed. Because the electrical solve and voltage feedback remain active in D_E, this result concerns the additional thermal/phase interior package. It does not test physics against a data-only network. Complete independent audit components remain in Supplement S17.

5.2 Second-cycle phase-gap reconstruction

Neither initialization establishes the complete phase-gap criterion against D_E under the three references and two readers. Across those conditions, relative phase-RMS reductions of E against D_E range from -4.23% to 1.23%; negative reduction means larger E error. The prescribed phase-gap experiment does not establish the claimed additional residual increment; individual favorable errors do not replace the complete criterion.

Seed 29 has phase-RMS error reductions of 1.10% to 1.23% and active-set error reductions of 1.30% to 1.46%. Seed 43 has phase-RMS error reductions of -4.23% to -4.18% and active-set error reductions of -5.25% to -5.21%. All three window guards jointly pass in 12/12 conditions. The phase-gain requirements still determine the complete A_w outcome; negative reductions indicate larger E errors.

Table 3. B1 errors inside W under the spatial reference and fine reader. All seven objects are included. Full-precision values for all references/readers, including their own denominators, remain in the complete tables.

Seed	Role	1000 S	Phase RMS	T (%)	Current (%)	Power (%)
29	E	8.92423	0.102408	1.14001	1.10488	1.11993
29	D_E	9.042	0.103548	1.13479	1.15561	1.17082
29	F	7.75023	0.121882	1.16888	8.96009	8.42207
43	E	8.0082	0.0906255	1.10747	0.931022	0.95043
43	D_E	7.61158	0.0869874	1.12689	0.996119	1.00989

Seed	Role	1000 S	Phase RMS	T (%)	Current (%)	Power (%)
43	F	7.75023	0.109148	1.11942	8.90024	8.38301
-	B_E	7.75023	0.127808	1.41376	9.08846	8.55694

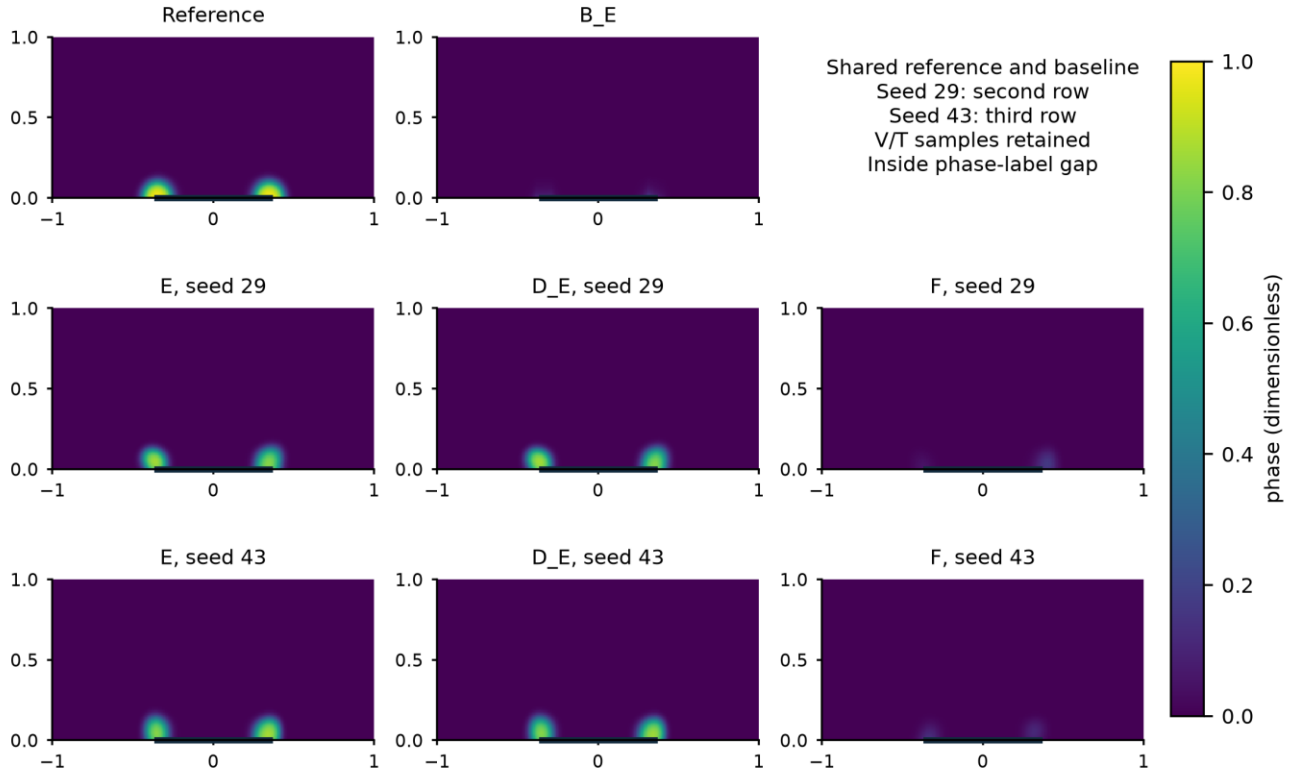
B1: phase at fixed $t=1.28$; native 240 x 120 fields

Figure 4. B1 phase at $t=1.28$, inside the withheld second cycle. Reference and shared B_E accompany E, D_E and F for both seeds. All panels use the same [0,1] color range. Native fine-grid fields provide a physical view, while the primary state and event criteria use the original coarse measure. T, local q, both predeclared times and signed-error companions are retained in the supplementary physical atlas.

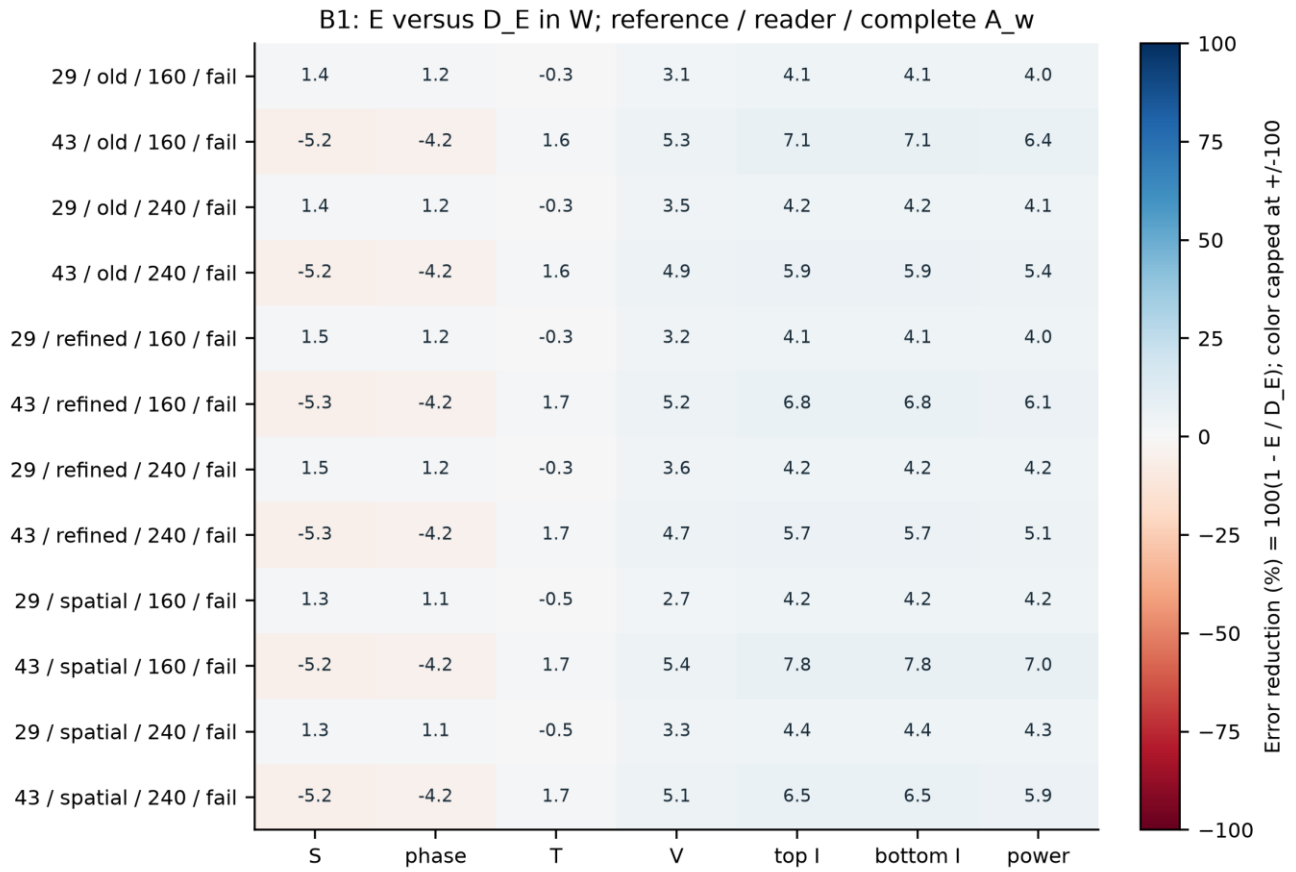


Figure 5. Window effects of E against same-parent D_E in all twelve seed/reference/reader conditions. Positive numbers mean smaller E error. Colors are capped at plus/minus 100% for legibility; annotations retain the actual values. The row label gives the complete A_w outcome, which also uses absolute floors and noninferiority. The rows are sensitivity checks on two independent initializations, not twelve independent experiments.

5.3 Whole-history costs and strict events

Outside-window noninferiority costs occur in 6 of the twelve primary comparison conditions. The affected quantities are top-current NRMSE, potential RMS, bottom-current NRMSE, power-trace NRMSE, with every condition retained in the comparison table. All seven B1 objects fail the complete strict two-cycle rule under every tested reference and reader.

Table 4. Primary B1 decisions in every reference/reader condition. Full A/B retain their original definitions; outside cost denotes at least one separately reported noninferiority violation. A favorable window result does not erase an outside or strict-event cost.

Reference	Reader	Seed	A w	Full A	Full B	Outside cost
old	coarse	29	False	False	False	False
old	coarse	43	False	False	False	True
old	fine	29	False	False	False	False
old	fine	43	False	False	False	True
refined	coarse	29	False	False	False	False
refined	coarse	43	False	False	False	True
refined	fine	29	False	False	False	False
refined	fine	43	False	False	False	True
spatial	coarse	29	False	False	False	False
spatial	coarse	43	False	False	False	True
spatial	fine	29	False	False	False	False
spatial	fine	43	False	False	False	True

Against the repaired soft PINN F, B1 E satisfies the same window reconstruction formula in 0/12 conditions and the original whole-history A/B rules in 0/12 and 6/12, respectively. Against the shared interpolant B_E, B1 E satisfies the same window reconstruction formula in 0/12 conditions and the original whole-history A/B rules in 0/12 and 10/12, respectively. These are reference/reader checks on the same two initializations. Their continuous effects and all failures remain in the complete tables; the primary attribution of the additional interior package still uses E versus D_E. For seed 29, the full device rule passes 0/6 checks against F and 4/6 against B_E. For seed 43, the full device rule passes 6/6 checks against F and 6/6 against B_E.

Table 5. Continuous B1 window effects across the three references and two readers. Each range summarizes six sensitivity conditions for one initialization; positive reduction means smaller E error. It does not establish the full window criterion.

Comparison	Seed	Phase reduction %	Set reduction %	Current reduction %	Power reduction %
E / D_E	29	1.10 to 1.23	1.30 to 1.46	4.06 to 4.39	3.98 to 4.35
E / D_E	43	-4.23 to -4.18	-5.25 to -5.21	5.70 to 7.79	5.14 to 7.04
E / F	29	15.98 to 16.93	-15.15 to -13.14	86.92 to 88.57	85.84 to 87.76
E / F	43	16.97 to 18.27	-3.33 to -1.01	88.69 to 90.58	87.70 to 89.86
E / B_E	29	19.87 to 20.69	-15.15 to -13.14	87.10 to 88.73	86.06 to 87.96
E / B_E	43	29.09 to 30.23	-3.33 to -1.01	88.92 to 90.78	87.95 to 90.07

The spatial-reference event records separate port and support behavior. Both E and D_E recover a second event, whereas both repaired F states and the shared B_E miss it. Second-cycle ROI recall is 0.7799/0.7738 for E/D_E at seed 29 and 0.8061/0.8010 at seed 43, below the 0.9 requirement. E second-event timing errors are 0.00450 and 0.00771, respectively. Thus substantially improved window ports and recovery of an event can coexist with failure of the phase and strict two-cycle criteria. The complete first/second-event records, including support extent, precision and recovery, are retained in Supplement S20.

All E/F and E/B_E comparisons, complete cycle metrics, signed device effects and separate full/outside normalizers are retained with the primary comparison. Phase observations after 2.02 support both the networks and the legal interpolant, so even a favorable result is offline reconstruction rather than online second-pulse prediction. The outcome has not changed the full-label D_E finding, historical E/F decisions, seeds, thresholds or reference definitions.

5.4 Bounded corrections and equation-wise diagnostics

Two later development questions retain the same synthetic physical object but have distinct parents and budgets. An eight-arm relative-residual/time-moment study does not establish its prescribed phase-gap increment; its eight arms share one parent and are not eight independent initializations. The combined RIM candidate improves phase RMS by 1.918% and set error by 2.202% over its matched D control, below the inherited practical requirements. These effects must not be pooled with the earlier B1 comparisons or interpreted as a cross-round learning curve. Supplement S21 retains every arm, raw audit and event record.

A separate fixed-parent study restricts neural N and spline S corrections to the dark interval $D=[1.36, 2.02]$, preserving the base temperature and port predictions and leaving phase unchanged outside D. General correction G is the direct unrestricted-support control. None establishes the prescribed completion increment or independent equation-wise qualification. N reduces the window set error by 34.21% while increasing phase RMS by 7.95%; unchanged ports therefore do not identify the hidden phase trajectory. These are changes in predictions, not new evidence of their accuracy.

For B0, the accepted N-Adam checkpoint, N-final and S-final, a subsequent zero-update diagnostic compares the same time nodes, spatial cells and boundary samples. At common 256-node temporal quadrature, N-final lowers H by 89.9073% while its raw phase-residual square becomes 10.3157 times B0. On a second spatial support these values are 80.7961% and 13.4327 times, respectively. The decomposition $H=J_{\phi}/75+J_T/48+5 B_{\phi}$ shows compensation between boundary and dynamic terms. The tested 128-to-256 temporal refinement changes each integrated component by less than 0.029%, whereas the spline boundary term depends strongly on spatial coverage. This is a directly observed objective/qualification mismatch, not an isolated proof of the complete optimization failure mechanism or a general claim about L-BFGS.

The temperature-implied phase trajectory under exact thermal balance violates phase-range and endpoint conditions on the checked support. Its thermal-square lower bound is about 21.615% of the same-support base value, which

does not exclude improvement within the allowed 105% thermal budget. The base itself belongs to that tolerance set. What remains unresolved is an attainable improvement satisfying the original support, range, endpoint and equation-wise conditions, followed separately by improved reference reconstruction and a neural-specific benefit. Supplement S22 preserves all controls and component records.

A supplementary fixed-temperature conditional evolution passes the two-step numerical checks and both paired thermal budgets. On the same 80 by 40 D-only development measure, phase RMS falls from 0.066944 to 0.020743, with the same direction in the original ROI. This gain comes with lower active-support recall: restricting the native reference indicator gives 0.974612 for B0 and 0.763383 for the fine conditional trajectory, while precision rises from 0.410817 to 0.890215. Thresholding after restriction separately gives recall 0.982021 to 0.776968. These D-only measures do not replace the original event rule or the historical W-window denominator. The left first-derivative seam RMS remains 0.497703 and the right value seam RMS is 0.020563. Thus the result informs a bounded feasibility question while establishing neither the original correction family's feasibility nor a neural-specific increment (Supplement S23).

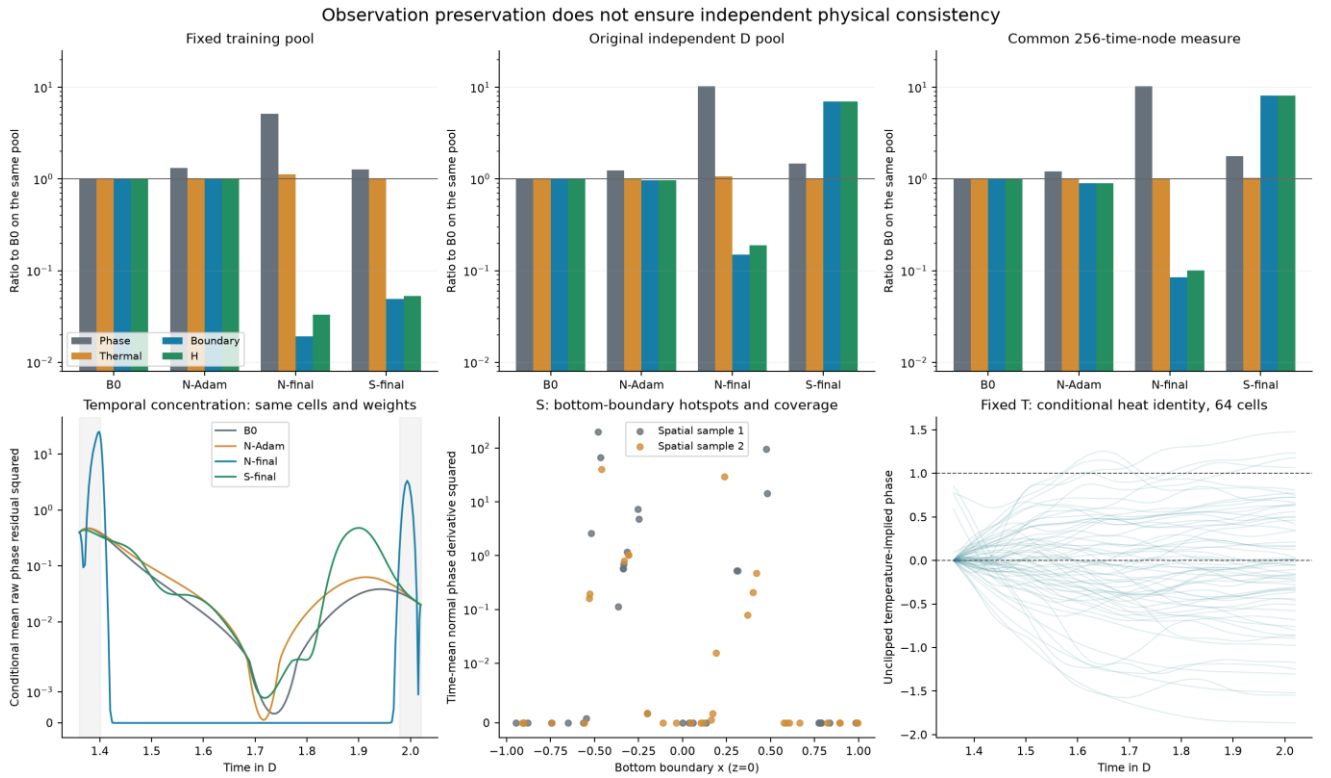


Figure 6. Saved-checkpoint diagnostic of scalar-objective reduction and separate physical components. Temporal refinement and spatial-support changes are distinct checks; values are not mixed across supports. This figure uses the completed zero-update diagnostic, not a new training run.

6. Physical interpretation and numerical scope

6.1 Pulse history and functional quantities

The references agree before the changed pulse to roundoff at matching resolution. Immediately before the second pulse, advancing it from 1.25 to 1.01 increases the reference ROI mean T from 0.006525 to 0.018746 and mean phase from 0.0004829 to 0.0039725. The second-event latency decreases from 0.2484 to 0.2168. These observations support history dependence in this numerical object but do not isolate thermal from phase mediation. Complete per-pulse states and event values remain in Supplement S5.

Device and event rankings differ. Historical shorter-protocol E passes both onset-timing tolerances under the original reference, but first-cycle recalls 0.866607 and 0.875507 remain below 0.9. F29 has the smaller second-cycle timing error. F43 has opposite signed pulse-energy errors, +0.00511674 and -0.00400099, yielding 0.25736% total energy error despite 3.09469% power-trajectory NRMSE. E also exhibits cancellation. Since projected current and power obey $P=UI$, their errors are related drive-weighted views; neither low energy error nor electrical balance certifies local heating or strict event reconstruction.

6.2 Reference and reader sensitivity

The original reference uses 160 by 80 cells and $dt=0.000625$; temporal refinement halves dt ; spatial refinement uses 240 by 120 at the halved step. All save the same 1001 times and keep the learned states fixed. Spatial fields are conservatively restricted for the primary state/event measure while ports remain native. Predictions have two separately tested electrical readers. These are finite perturbation tests, not a demonstrated convergence order or continuum-error estimate.

Across the historical coarse-reader comparisons, temporal refinement changes no A/B decision; spatial refinement changes three: original/29 E/B_E loses A, shorter/43 E/F loses A, and original/43 E/B_E gains B. The latter B crossing disappears with the finer reader. Section 4.2 additionally retains the continuous B_E reversals under original/time references. The complete E/F device criterion remains retained in every tested condition. Supplement S9 and S14-S16 give the complete margins, sufficient perturbation bounds and restriction conventions.

A later gated continuation crosses the strict rule only under the time-refined reference and loses it under spatial refinement as first-cycle recall changes from 0.902063024 to 0.897526906. Ordinary and gated phase-head extensions do not establish a stable matched increment over unchanged continuation. All six negative continuation states are retained in Supplement S10-S13 and the separately indexed array extension. These limits prevent attributing the core effect to a demonstrated new gate, extra capacity or arbitrary-grid robustness.

6.3 What the experiment identifies

The E/F contrast identifies the tested constraint-training configuration. Initial V, active parameters, exact versus finite enforcement, enforcement times and gradient maps differ; the experiment does not isolate one implicit-gradient pathway. E/D_E controls that distinction more tightly for the additional interior residuals, but its conclusion is conditional on the information regime and frozen optimization recipe. Each information regime rebuilds its own parents and calibration scales. The comparison between regimes therefore describes the observed pattern of increments without holding every training mediator fixed. Equal update limits do not give equal numerical work, and no acceleration over a conventional forward solver is claimed.

The domain has a localized electrode and a closed electric-thermal-phase feedback chain. Its dimensionless coefficients, single phase variable and idealized geometry still require a separate material-consistent calibration and independent validation before a device-material claim. The initial-boundary incompatibility and shared numerical discretization likewise remain limitations. A stronger algorithmic or material claim would need a separately designed experiment rather than relabeling the present controls.

An auxiliary test on the distinct, fitted lumped VO₂ neuristor model of Qiu et al. [19] shows why port accuracy and state consistency require separate checks (Supplement S25). From the same 197 finite voltage observations, PCHIP and cubic-spline reconstructions at 12.5 V give temperature RMS errors of 0.654 and 0.761 K against the frozen numerical source, while a native-voltage control gives 0.0189 K. Replaying the constitutive history leaves circuit-closure RMS errors of 223.27 and 243.73 μA ; a closer constitutive current in another role does not remove that closure defect. The storage partition and closure identities in S25 explain these quantities without assigning causal percentages. This auxiliary numerical model is neither a calibration of the two-dimensional object nor experimental temperature validation; the matched joint prototype in S25 fails its prospective increment: eliminated, soft and spline configurations give joint current RMS errors of 116.82, 102.53 and 171.69 μA , with different heat and observation tradeoffs.

7. Conclusions and availability

Training-time electrical elimination with consistent local Joule deposition produces a matched device-reconstruction benefit over the original F soft-PINN configuration after both receive the same final electrical repair. All four full-label pairs retain the complete device criterion under the declared reference and reader changes. Strong interpolation supplies a continuous reversal and a reader-sensitive advantage decision, limiting the scope of that benefit. The stronger mixed-measure F_{cov} comparison retains the E device criterion in 12/12 shorter-protocol sensitivity conditions; this outcome further bounds the configuration-level claim.

Matched full-label and phase-gap controls do not establish the expected additional dynamic-residual increment. The bounded relative-residual and observation-preserving studies retain this limitation; lower scalar loss and unchanged electrical ports do not establish a qualified internal phase reconstruction. Strict two-cycle usability remains

unestablished. The evidence separates a configuration-level benefit from unresolved algorithmic, generalization and material claims.

Data, code and author declarations

Curated manuscript and evidence releases are publicly available in the project repository. The [circuit V/I scoring subset at commit 709b10f](#) includes ten saved numerical source-voltage/current records, finite voltage observations, locked PCHIP and cubic-spline predictions and polynomial coefficients, fixed scoring code and complete result tables. Its entry supports independent saved-array rescoring; the recorded isolated-directory rerun reproduced the circuit results exactly. This subset does not provide new ODE/PDE integration, checkpoint inference, neural AD residual evaluation or retraining. The published conditional-thermal package at commit d068bdf additionally retains the forty locked thermal responses, history replays and their scoring inputs; its saved-array entry is separate from the original conditional integrations. The present revision and any new joint-reconstruction outputs are supplied locally for review and are not claimed to be publicly deposited. A standalone numerical package retains 72 core, 18 historical continuation and 42 B1 reference/reader records, all eight moment arms, the correction controls, the saved conditional-evolution and residual arrays, and all twelve new F_{cov} reference/reader records. These counts are not independent replications. Its entry was actually run in a separate directory with access to the original repository denied; missing required inputs fail instead of falling back to that repository. The core reproduction checked 8,868 scalar, identity and Boolean entries at the original $\text{rtol}=2\text{e-}10$ and $\text{atol}=2\text{e-}12$, with exact categorical agreement. Saved-array rescoring, saved-residual reaggregation, new checkpoint inference and retraining are separate capabilities. Reaggregating archived AD residual values is not a new computation of neural AD residuals. No independent external retraining is claimed. The accompanying evidence map, package guide and per-scope verification records identify the complete inputs and actual reproduction level.

The separate full prediction/reference field arrays and checkpoints for the two-dimensional PINN study remain local and have no public dataset DOI. Public or controlled reviewer access, appropriate distribution permission and persistent archiving for that full-field package require author approval; publication of the circuit subset does not close the full-paper access gap (P03). AI-assisted preparation: OpenAI Codex assisted with research-code preparation, numerical execution, source checking, analysis, figures and drafting. Human authors must verify the content and take responsibility for interpretation, disclosures and submission. Funding, author contributions, competing interests, affiliations and final author approval remain to be supplied truthfully. This numerical study generates no experimental or human-subject data; cited third-party device measurements retain their separate source and rights.

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